

EE 3201.302

Lab 10 RLC Transient Analysis

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I. Abstract

The goal of this lab is to understand how various RLC circuits behave when different inductor values are used and how these changes impact damping. In this experiment, PSpice models were created for a series RL circuit, a series RC circuit, and a series RLC circuit to compare simulated results with real-world measurements taken using the AD2 and Waveforms. By capturing both the step and natural responses through circuit simulation over an appropriate time and reaching steady state, a transient known as the natural response was observed. Through correctly wiring the components and running the simulations to study the fundamental voltage and current characteristics, it was found that the theoretical predictions closely matched the experimental lab results.

II. Introduction

Transient analysis of circuits is necessary whenever voltages or currents experience sudden changes. The most common example of transient behavior occurs when switches are either turned on or off within a circuit. Energizing a previously inactive circuit by turning a switch on generates a transient called the step response, while turning a switch off in a circuit that has reached steady state produces a transient known as the natural response. To observe both the step and natural responses in PSpice, a Time Domain analysis was conducted using a pulsed voltage source — first applying voltage to activate the circuit, then, after enough time for transients to settle, turning it off. Construction of the three series circuits (RL, RC, and RLC) was accomplished in PSpice by utilizing the component library, enabling accurate comparisons to the circuits physically built during the lab.

III. Procedure

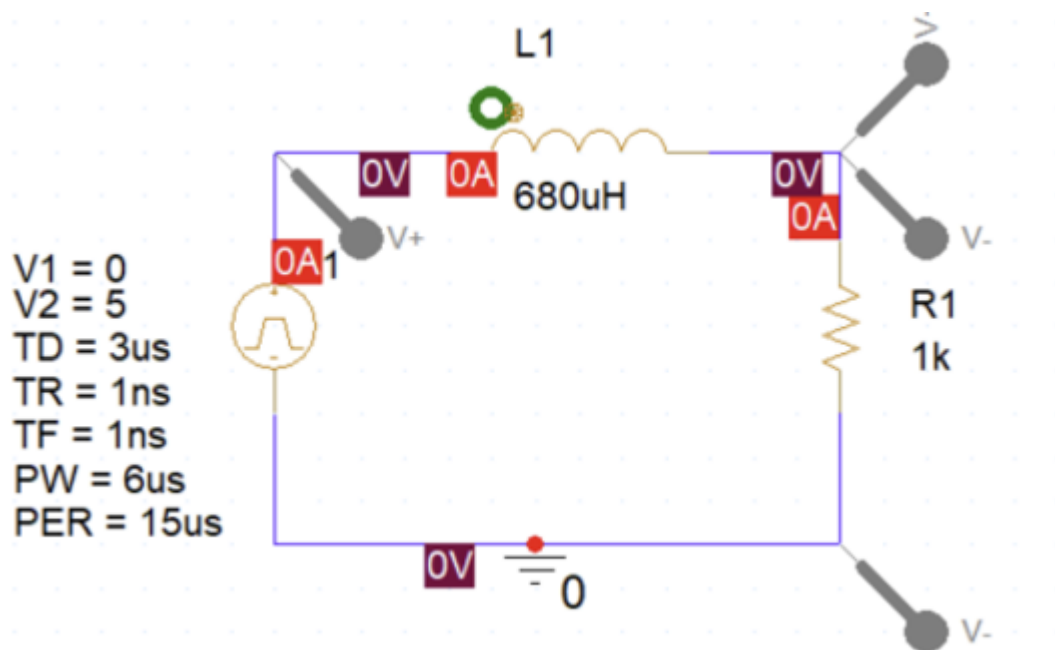


Figure 3.1: RL Circuit

1. Construct the RL circuit shown in Figure 3.1, using a $1\text{k}\Omega$ resistor and a $680\mu\text{H}$ inductor.
2. Connect the AD2 device to the circuit and configure Waveforms to supply a 5V peak-to-peak square sine wave with a 2.5V offset.
3. Run the simulation in Waveforms and create a math channel to display the current response across the resistor by measuring the voltage across it. Use the relation $I = V/R$ to calculate the current.
4. Compare the experimental results with the prelab predictions to determine if the current and voltage responses match for the RL circuit.

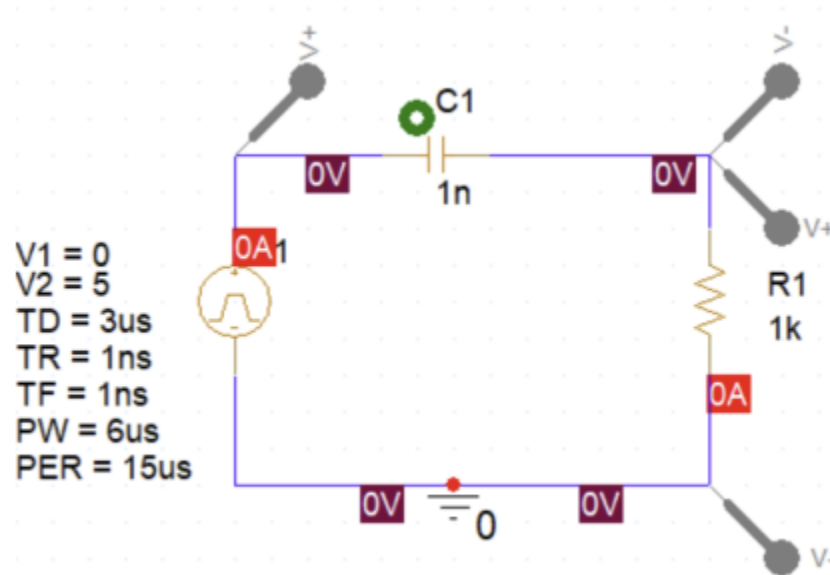


Figure 3.2: RC Circuit

5. Construct the RC circuit shown in Figure 3.2, using a $1\text{k}\Omega$ resistor and a 1nF capacitor.
6. Compare the current and voltage responses obtained in the lab with the prelab results to verify if they are consistent for the RC circuit.

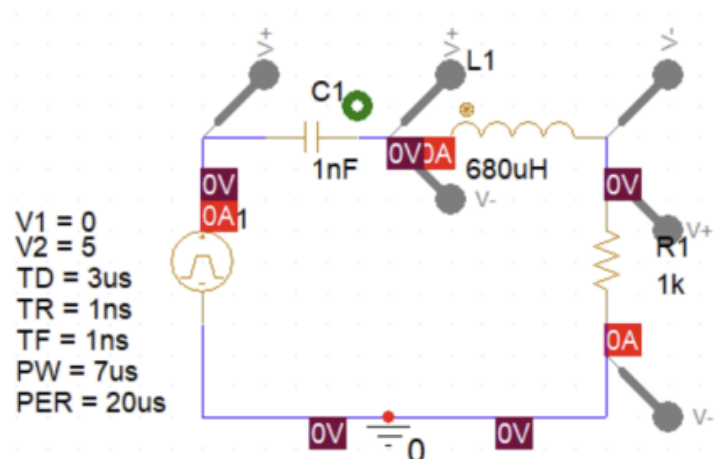


Figure 3.3: RLC Circuit

7. Build the RLC circuit shown in Figure 3.3, using a $1\text{k}\Omega$ resistor, a 1nF capacitor, and a $680\mu\text{H}$ inductor.
8. Set up the reference channels in Waveforms to display the voltage and current responses for all three components — the inductor, resistor, and capacitor — along with the current through the resistor.
9. Compare the measured current and voltage responses with the prelab results for the RLC circuit. Identify whether the circuit behavior is underdamped, critically damped, or overdamped.
10. Replace the inductor with a different value and capture the new simulated waveform to observe a different damping behavior. Determine whether the circuit is now underdamped, critically damped, or overdamped. Repeat this step again with another inductor value to obtain one example of each damping condition.
11. Compare the simulated waveforms to the predictions made during the prelab for the RLC circuit.

IV. Pre-lab Results

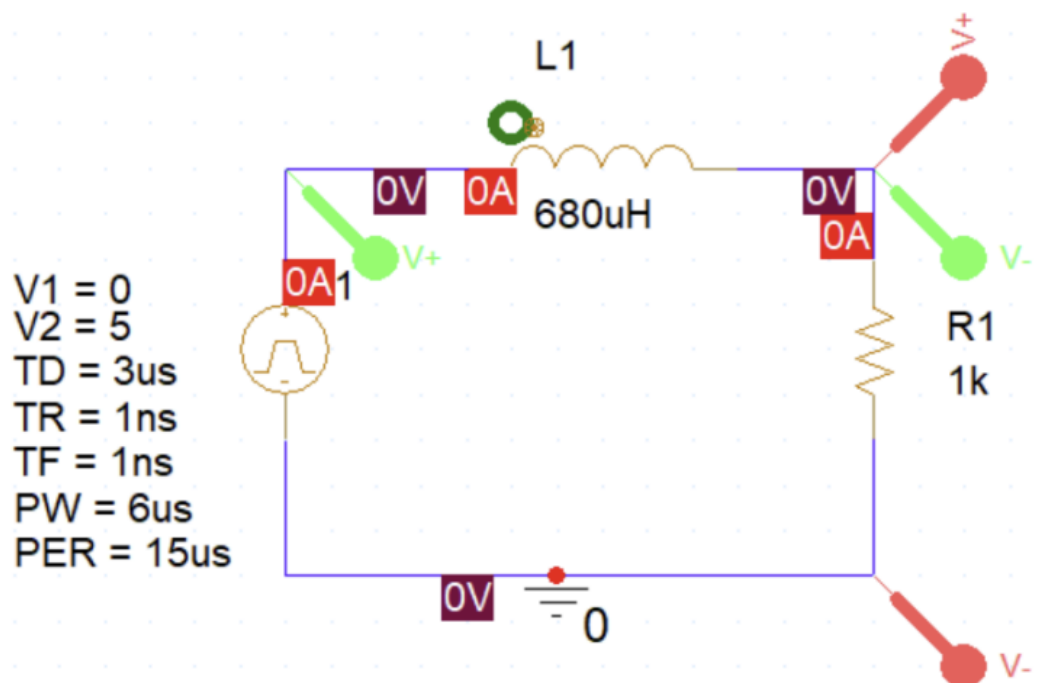


Figure 3.1: RL Circuit with Voltage and Current measurement

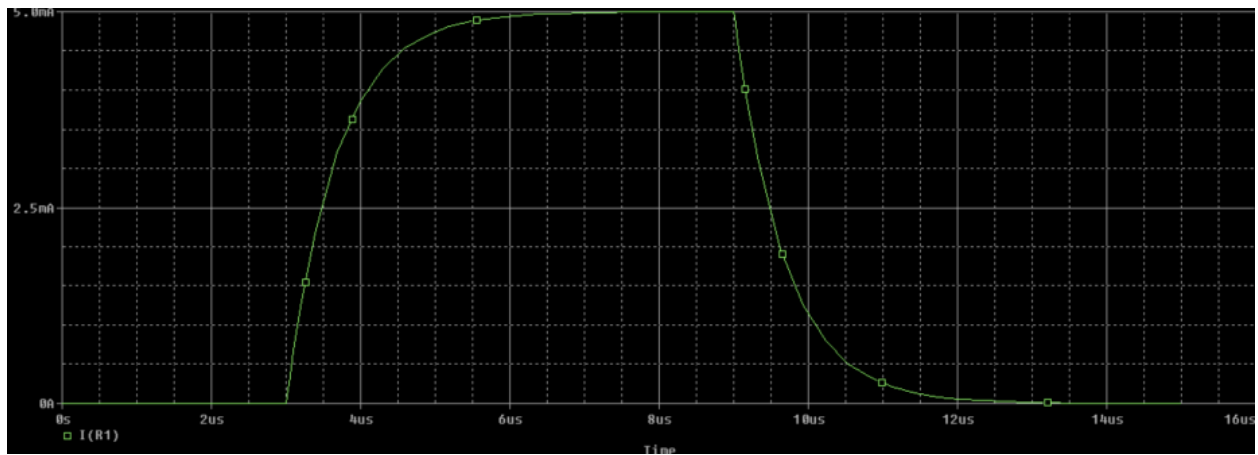


Figure 3.2: Natural Current Response of RL circuit

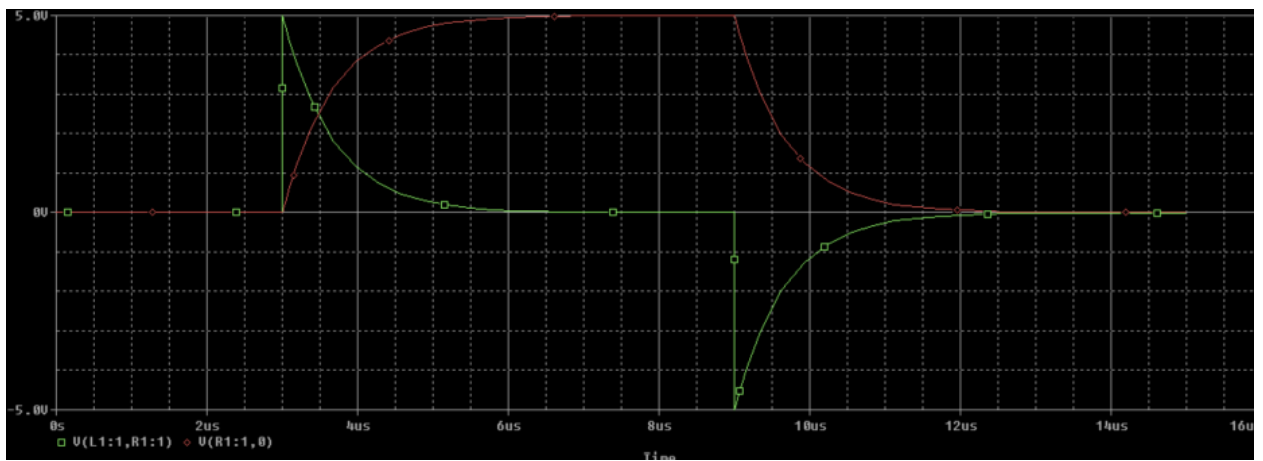


Figure 3.3: Transient Voltage Response of RL circuit

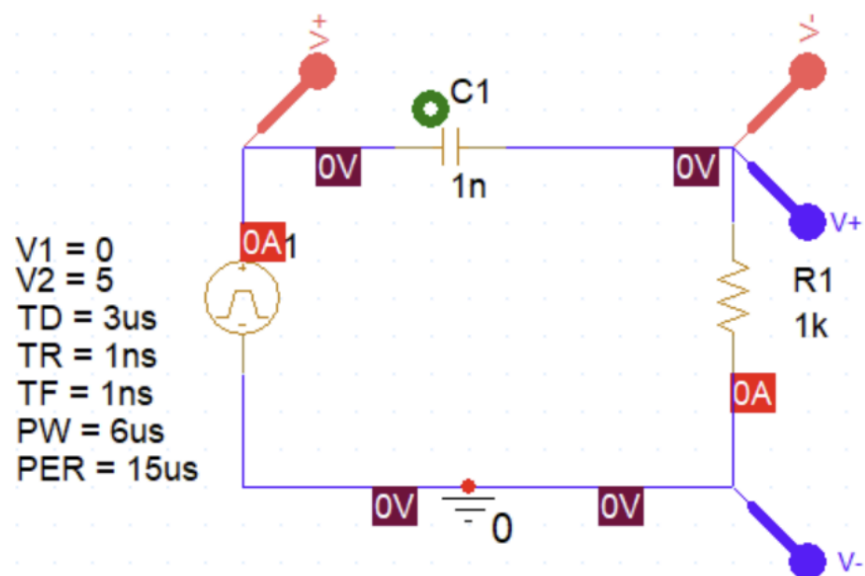


Figure 3.4: RC circuit with Voltage and Current measurement

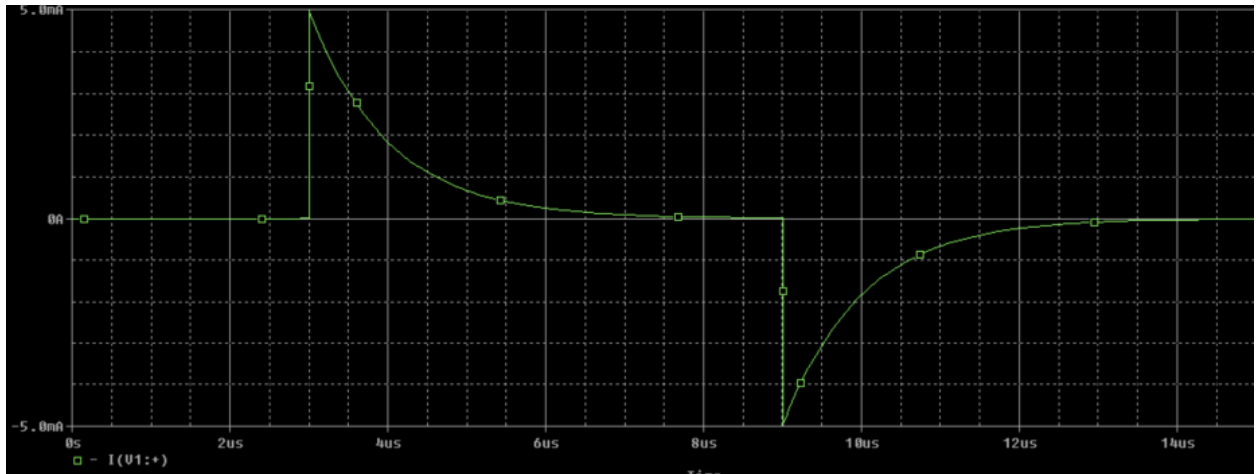


Figure 3.5: Natural Current Response of RC circuit

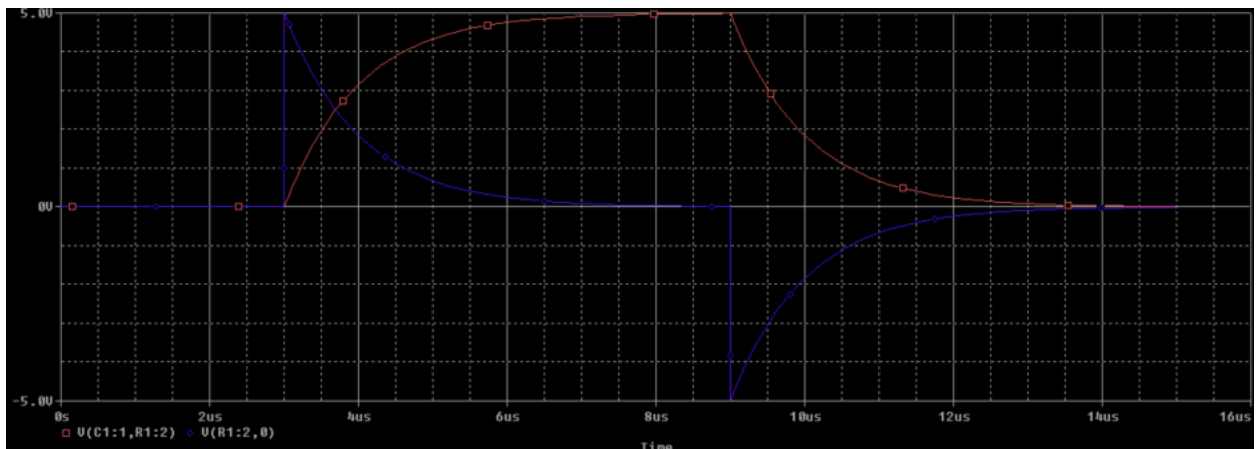


Figure 3.6: Transient Voltage Response of RC circuit

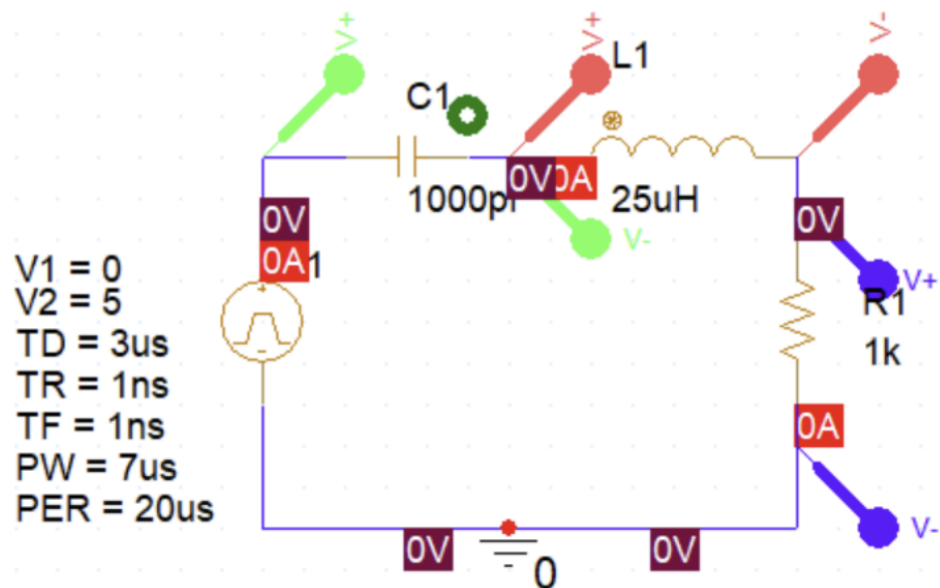


Figure 3.7: RLC circuit with Voltage and Current measurements

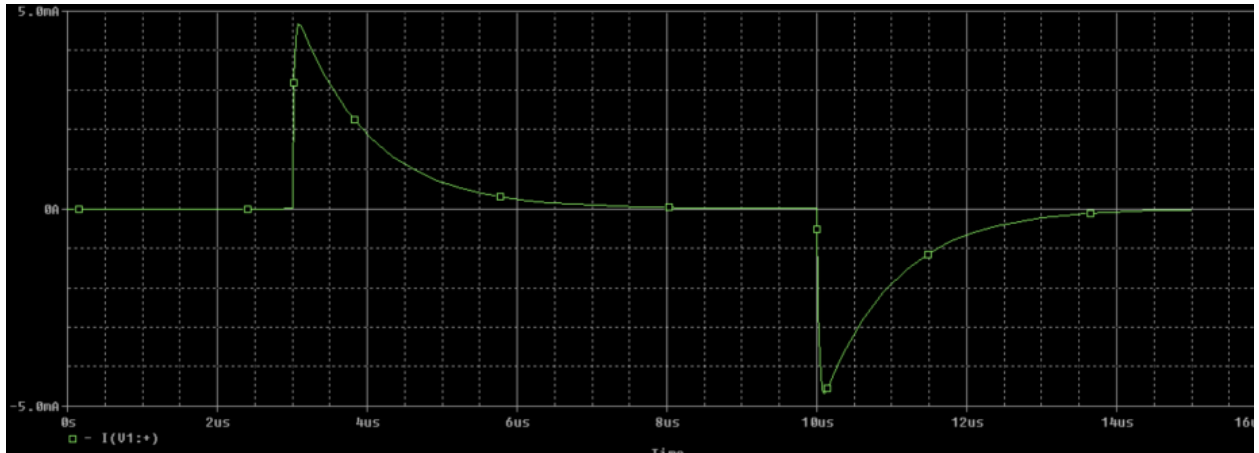


Figure 3.8: RLC circuit Transient Current Response

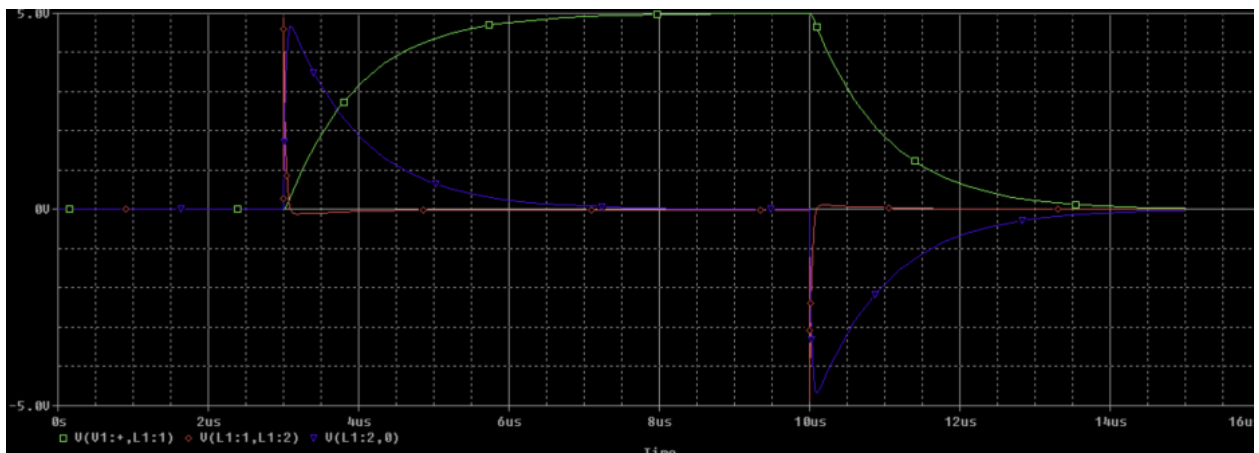


Figure 3.9: RLC circuit Transient Voltage Response

V. Experimental Results

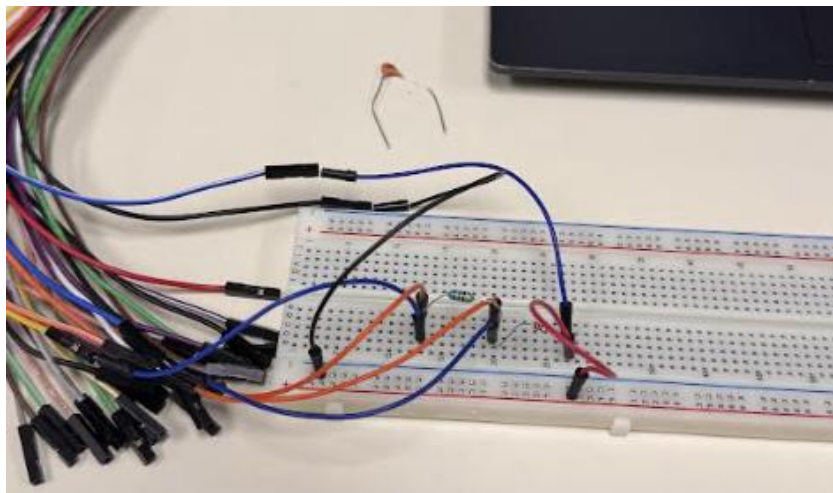


Figure 5.1: Physical RL Circuit



Figure 5.2: AD2 Waveforms for Voltage and Current response of RL circuit

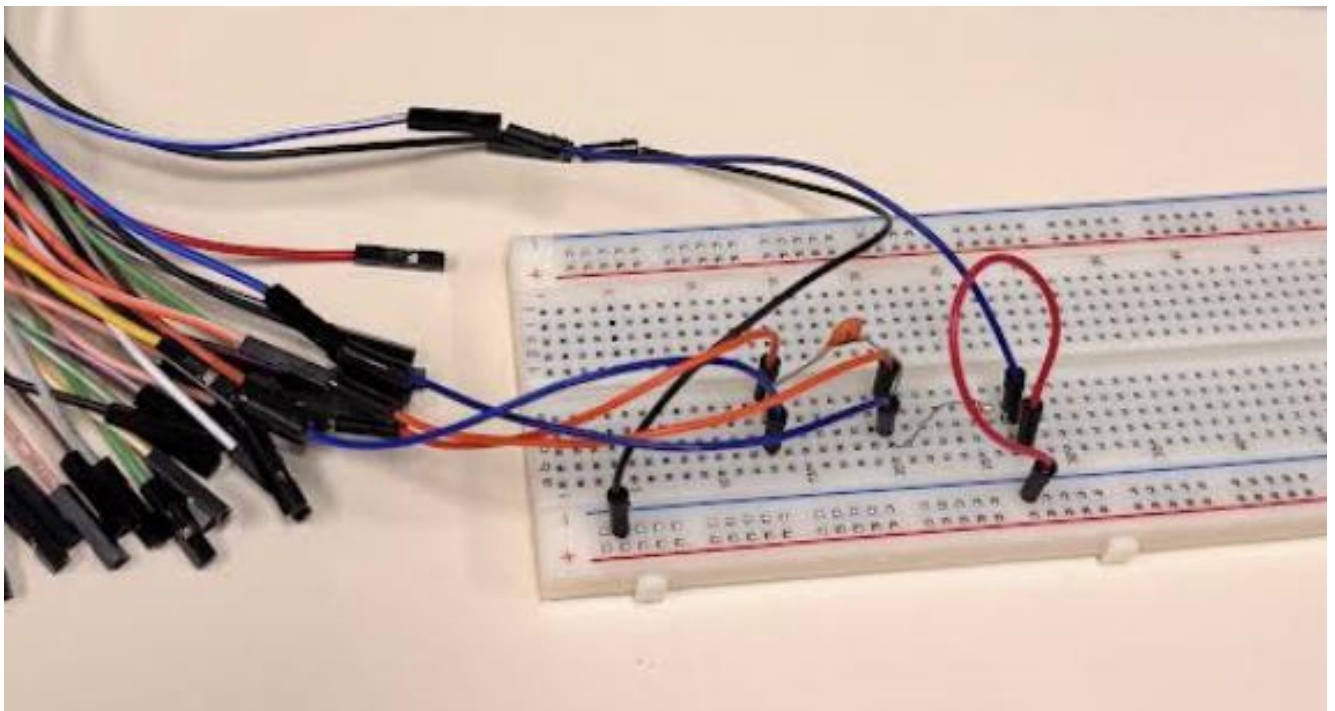


Figure 5.3: Physical RC Circuit



Figure 5.4: AD2 Waveforms for Voltage and Current Response of RC circuit



Figure 5.5: Zoomed in Waveform for RC circuit

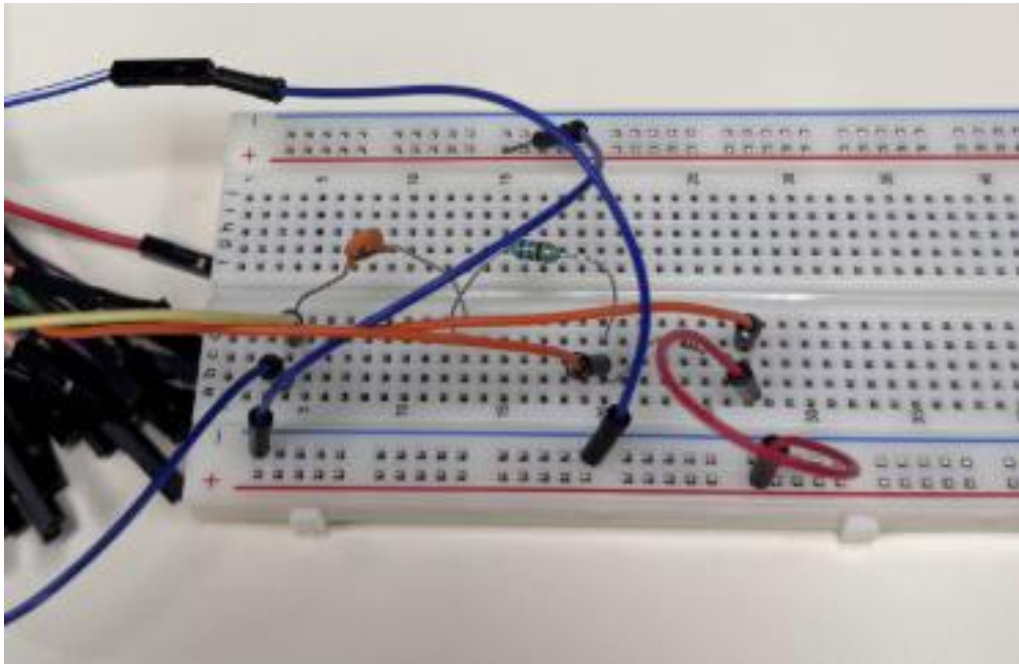


Figure 5.6: Physical RLC Circuit



Figure 5.7: AD2 Waveform of Natural Response RLC Circuit



Figure 5.7: AD2 Waveform of RLC Circuit over Capacitor



Figure 5.8 AD2 Waveform of RLC Circuit over Inductor



Figure 5.9 AD2 Waveform of RLC Circuit over Resistor

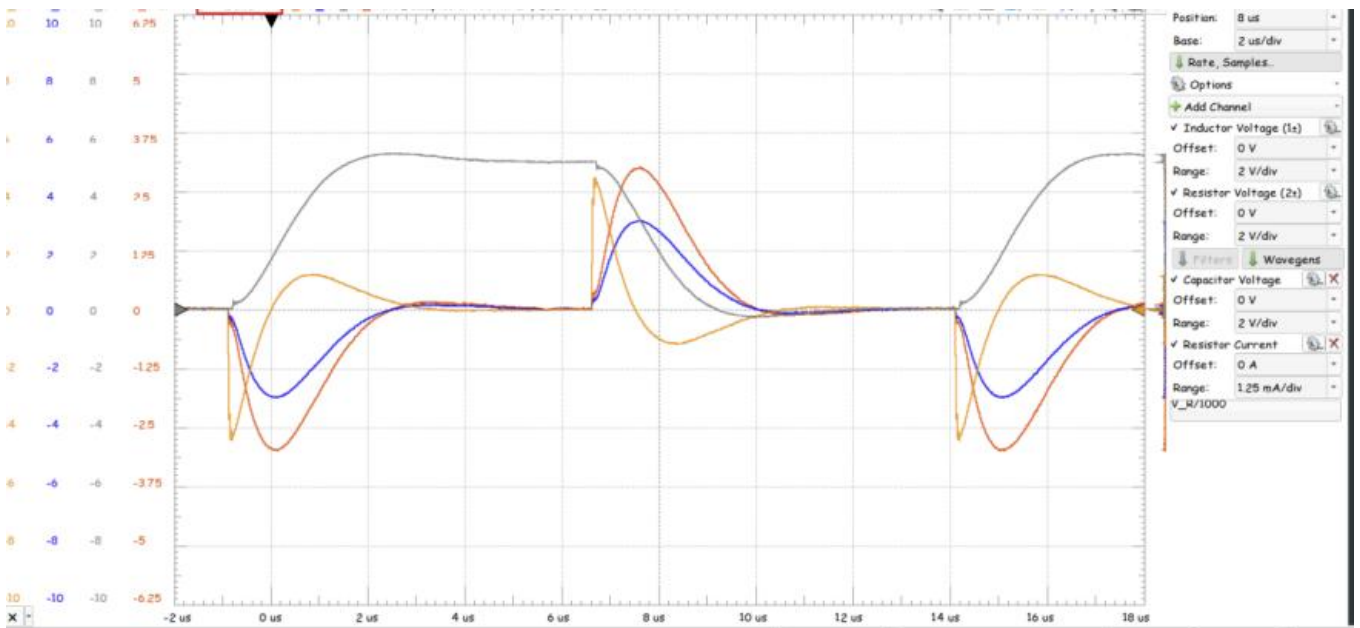


Figure 5.10: Critically Dampened RLC Circuit

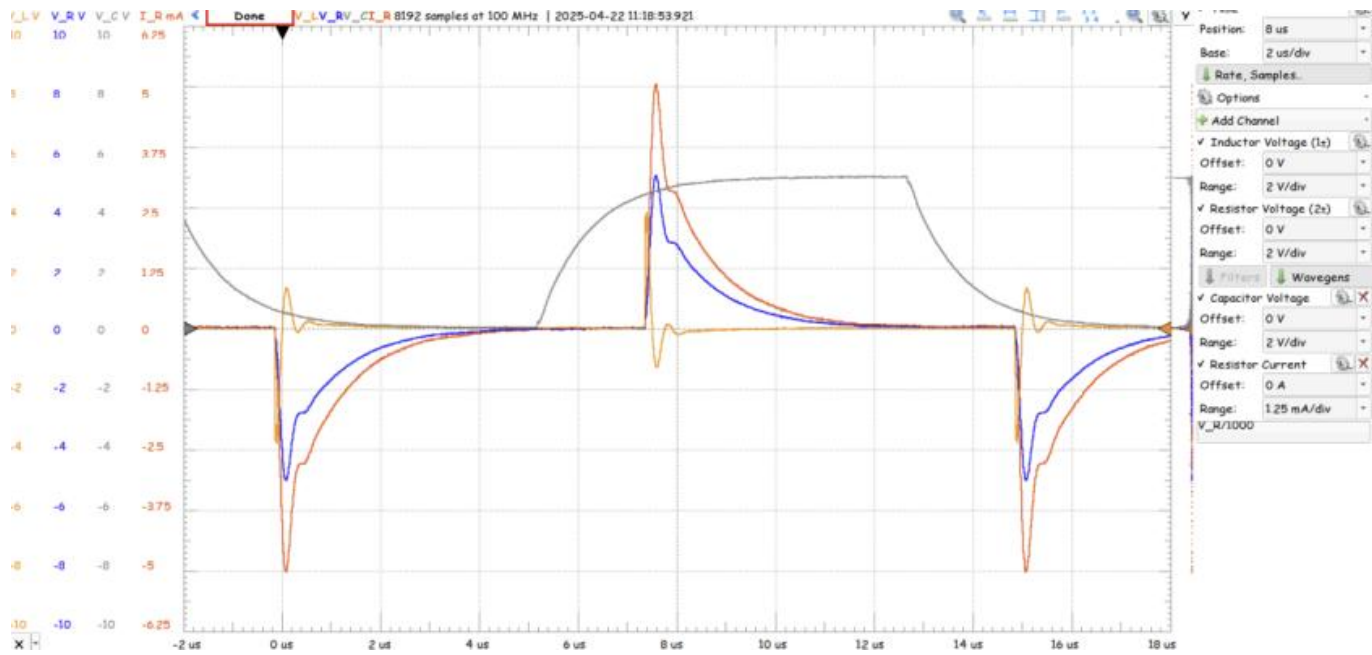


Figure 5.11: Overdamped RLC Circuit

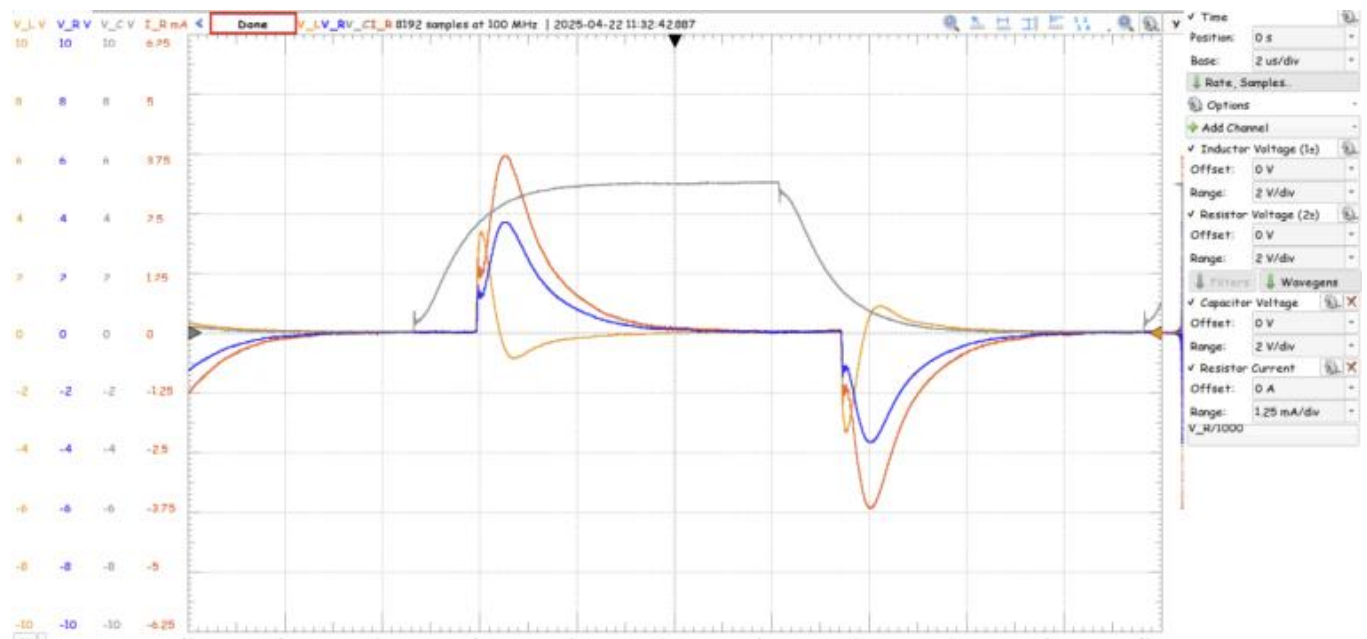


Figure 5.12: Underdamped RLC Circuit

VI. Discussion and Conclusions

Overdamped, critically damped, and underdamped behaviors describe how the resistor, inductor, and capacitor interact within a system, and how the system responds to a disturbance or input signal. Damping is determined by how much energy the resistor dissipates compared to the energy stored in the inductor or capacitor. An underdamped response occurs when resistance is low, causing the system to oscillate while gradually settling down, with decreasing oscillation amplitude over time. A critically damped response represents the ideal case where the system returns to steady state as quickly as possible without oscillating. In contrast, an overdamped response occurs when resistance is too high, leading to a slower return to equilibrium without any oscillations.

For the RLC circuit tested, the overdamped condition occurred with the $56\mu\text{H}$ inductor, the critically damped condition with the $680\mu\text{H}$ inductor, and the underdamped condition with the 1mH inductor. These findings were consistent with the simulated results from the prelab, where the overdamped circuit used a $25\mu\text{H}$ inductor, the critically damped circuit used a $250\mu\text{H}$ inductor, and the underdamped circuit used a $500\mu\text{H}$ inductor. The consistency between the lab and simulation results makes sense, given the expected behavior based on inductor values. Inductors are designed to smooth out changes in current; however, using too large of an inductor can lead to overdamping, where the voltage and current take too long to stabilize. On the other hand, using too small of an inductor can cause underdamping, resulting in oscillations. A properly selected inductor value that achieves critical damping is optimal for balancing circuit efficiency with speed, which is highly valuable in real-world applications requiring fast system responses.

This lab successfully demonstrated how RL, RC, and RLC series circuits behave over time and how the type of damping affects their responses at different frequencies. Both step and natural responses were analyzed, and the simulations closely matched the hands-on results obtained using the AD2 and Waveforms, following the lab manual's procedures. PSpice modeling proved effective in predicting circuit behavior, and building circuits based on schematic diagrams from the component library allowed for accurate comparison between simulated and real-world results. Overall, the lab provided valuable insight into transient circuit behavior and the importance of component selection in designing stable and efficient systems.